

# [P5] Lower bounds for $g(r)$ at every $r$ : $3r - 3$ , and $3r - 2$ from $r = 16$

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## Abstract

Let  $g(r)$  be the least number of hyperedges in an intersecting  $r$ -uniform hypergraph with transversal number  $r$ . Sivashankar [2] proved  $g(r) \geq 3r - 4$  and asked whether his residual bound  $4\tau \leq q + r + 4$  can be improved to  $+3$ . We give a proposed proof that it can (Theorem 1), so  $g(r) \geq 3r - 3$  for every  $r$ ; this is attained at  $r = 3$  and  $r = 4$ . We then exclude equality for every  $r \geq 16$  (Theorem 2), so  $g(r) \geq 3r - 2$  there. Every finite step is checked exhaustively by a named script. The asymptotic bound  $g(r) \geq (3.1108 - o(1))r$  is in the companion paper [P6].

## 1. Setting

### 1.1 The problem

A hypergraph  $H$  is **intersecting** if any two hyperedges meet, and  $\tau(H)$  is the least number of vertices meeting every hyperedge. Then

$$g(r) = \min\{|E(H)| : H \text{ intersecting, } r\text{-uniform, } \tau(H) = r\}.$$

The problem goes back to Erdős and Lovász [1], and Kahn [3] proved  $g(r) = O(r)$ . The known values are  $g(3) = 6$  and  $g(4) = 9$  [5],  $g(5) = 13$  [4] and  $g(6) = 17$  ([P1]–[P4] of this set). Against  $3r$  these are  $3r - 3$ ,  $3r - 3$ ,  $3r - 2$  and  $3r - 1$ .

**The reduction of [2].** Repeatedly take a vertex of degree at least four and delete the hyperedges through it. If this takes  $k$  steps and leaves a family  $J$  of  $q$  hyperedges and maximum degree at most three, then

$$|E(H)| \geq q + 4k, \quad r \leq k + \tau(J).$$

So a bound  $4\tau(J) \leq q + r + C_0$  gives  $|E(H)| \geq 3r - C_0$ . Theorem 1 is this bound with  $C_0 = 3$ .

### 1.2 The remainder J

We use the dual picture: each hyperedge of  $J$  is a **point**, and each vertex is a **block**, the set of points containing it. Since  $J$  is intersecting,  $r$ -uniform and of maximum degree three:

- every block has at most three points, and every pair of points lies in some block;
- every point lies in exactly  $r$  blocks (two vertices with the same block count twice).

A point  $A$  meets the other  $q - 1$  points through its  $r$  blocks, each of which reaches at most two of them. So

$$q - 1 \leq \sum_{v \in A} (\deg_J v - 1) \leq 2r. \tag{1.1}$$

**Triples,  $W$  and  $U$ .** Fix a maximum family of pairwise disjoint 3-point blocks  $M_1, \dots, M_t$ . Let  $W$  be their union ( $3t$  points) and  $U$  the other  $u$  points, so  $q = 3t + u$ . No 3-point block lies inside  $U$ , by maximality.

**The natural cover.** One vertex for each  $M_i$ , one common vertex for each pair of a pairing of  $U$ , and one more vertex if  $u$  is odd, cover  $J$ . So

$$\tau(J) \leq t + \lceil u/2 \rceil. \quad (1.2)$$

**Witnesses.** For each pair  $x, y \in U$  fix one common vertex  $w_{xy}$ , the *witness* of the pair. Its block is  $\{x, y\}$  or  $\{x, y, z\}$  with  $z \in W$ . Different pairs have different witnesses, since a repeated witness would give a block of four points or a 3-point block inside  $U$ .

**Local graphs.** For each  $i$ , the *local graph*  $F_i$  is the graph on  $U$  with an edge  $xy$  of colour  $z \in M_i$  whenever the witness block is  $\{x, y, z\}$ . Two disjoint edges of  $F_i$  always have the same colour, since otherwise their blocks would replace  $M_i$  by two disjoint triples.

**Notation used throughout.**

- For one colour:  $e$  = its number of edges,  $s$  = its support (vertices it touches).
- $S_i$  = sum of the three colour supports of  $F_i$ , and  $S = \sum_i S_i$ .
- $S_x$  = number of points of  $W$  reached by the witnesses at  $x \in U$ ; then  $S = \sum_x S_x$ .
- $d = 2r - q + 1$ , the slack in (1.1).

## 2. Theorem 1: the sharp residual bound

**Theorem 1 (proposed).** *Every intersecting  $r$ -uniform family  $J$  with  $q$  hyperedges and maximum degree at most three satisfies  $4\tau(J) \leq q + r + 3$ .*

**Corollary 1.**  $g(r) \geq 3r - 3$  for every  $r \geq 1$ .

*Proof.* The reduction of §1.1 with Theorem 1.  $\square$

### 2.1 Capacity and local support

**Capacity at  $U$ .** The witnesses use exactly  $u(u-1)$  of the  $ur$  incidences of points of  $U$ . Each of the other  $ur - u(u-1)$  incidences reaches at most two points of  $W$ , and all  $3tu$  pairs between  $U$  and  $W$  must be met. Hence

$$3tu - S \leq 2(ur - u(u-1)). \quad (2.1)$$

**Capacity of a colour.** Fix a colour  $z \in M_i$  with  $e$  edges and support  $s$ . In the point  $z$ , the  $e$  witnesses cover  $s$  points of  $U$ , the vertex of  $M_i$  covers its two other points, and the remaining  $q - 3 - s$  points need the other  $r - e - 1$  vertices, at most two each. So

$$2e - s \leq 2r - q + 1 = d. \quad (2.2)$$

**Lemma 1 ([2] for the first bound).** *Let  $F$  be a simple graph on  $u$  vertices with at most three colours, in which disjoint edges have the same colour. Then  $S_F \leq \max(12, u + 4)$ . If every colour has  $2e - s \leq d$ , then  $S_F \leq \max(12, u + 2, d + 8)$ .*

*Proof.* Suppose first that some colour has two disjoint edges, with endpoint set  $Y$  of four vertices. Every edge of another colour meets both, so it lies in  $Y$ .

- If no other colour is present,  $S_F \leq u$ .
- If the other-colour edges have no common vertex, every first-colour edge meets all of them, so it lies in  $Y$  too, and  $S_F \leq 12$ .

- Otherwise there are at most two other-colour edges, sharing a vertex, contributing at most four:  $S_F \leq u + 4$ . Fix one of them,  $g$ . Every first-colour edge meets  $g$ , so  $e \geq s - 2$ , and  $2e - s \leq d$  gives  $s \leq d + 4$ , hence  $S_F \leq d + 8$ .

Suppose next that no colour has two disjoint edges. Then each colour is a star or part of a triangle. If all supports are at most four,  $S_F \leq 12$ . Otherwise one colour is a star with at least four leaves; every other-colour edge contains its centre, and simplicity gives  $S_F \leq u + 2$ .  $\square$

## 2.2 A preliminary bound

**Lemma 2 ([2]).**  $r \geq u + t - 2$ .

*Proof.* Put  $\alpha = r - u - t + 2$ . If  $u = 0$ , (1.1) gives  $\alpha \geq (r + 5)/3 > 0$ . Otherwise (1.1), (2.1) and Lemma 1, with  $K = \max(12, u + 4)$ , give

$$t \leq u - 3 + 2\alpha, \quad \alpha \geq 1 - \frac{(K - u)t}{2u}.$$

A negative integer  $\alpha$  would give  $t \leq u - 5$  and then  $\alpha > -1$ . For  $K = u + 4$  the bound is  $-1 + 10/u$ ; for  $K = 12$  with  $u \leq 8$  it needs  $u^2 - 13u + 60 > 0$ , whose discriminant is negative.  $\square$

## 2.3 The odd equality case

By (1.2) and Lemma 2:

- if  $u$  is even,  $4\tau \leq 4t + 2u = q + (t + u) \leq q + r + 2$ ;
- if  $u$  is odd,  $4\tau \leq q + (t + u) + 2$ .

So Theorem 1 follows once we show: *if  $u$  is odd then  $r \geq u + t - 1$* . Suppose  $u$  is odd and  $r = u + t - 2$ . Then  $d = u - t - 3 \geq 0$ , and (2.1) becomes

$$S \geq u(t + 2). \tag{2.3}$$

*Large  $u$ .* If  $u \geq 11$ , (2.3) and Lemma 1 give  $t \geq u/2$ . Then  $d + 8 \leq u + 2$  and  $12 \leq u + 2$ , so every  $S_i \leq u + 2$  and  $u(t + 2) \leq t(u + 2)$ , that is  $u \leq t$ , against  $t \leq u - 3$ .

*Small  $u$ .* For odd  $u \leq 9$ , the conditions  $0 \leq t \leq u - 3$  and  $u(t + 2) \leq t \max(12, u + 4)$  leave exactly

$$(u, t) \in \{(5, 2), (7, 3), (7, 4), (9, 5), (9, 6)\},$$

with  $d = 0, 1, 0, 1, 0$  respectively.

**Lemma 3 (small excess).** *If every colour has  $2e - s \leq 1$  and  $F$  has more than one colour, then  $S_F \leq 12$ ,  $S_F \neq 11$ , and  $S_F = 12$  only when  $F$  is a properly three-coloured  $K_4$ . If also  $u \leq 9$ , the same holds when  $F$  has one colour.*

*Proof.* A connected component on  $v \geq 2$  vertices has  $2e - v \geq v - 2$ , so each colour is a matching, or one path on three vertices plus a matching.

- A colour with a matching of size three leaves the other colours empty, so  $F$  has one colour and  $S_F \leq u \leq 9$ .
- A colour of support five is  $P_3$  plus an edge; the other colours are then confined to two edges, and  $S_F \leq 9$ .

- Otherwise every support is at most four. Support four forces two disjoint edges, and two cross-intersecting two-edge matchings are two perfect matchings of one  $K_4$ ; a third colour must be the third.  $\square$

The cases  $d = 1$ .

- $(9, 5)$ : (2.3) asks  $S \geq 63$ , and Lemma 3 allows 60.
- $(7, 3)$ : (2.3) asks 35, so all three  $S_i$  are 12, that is, three edge-disjoint copies of  $K_4$  on seven vertices. Two such copies meet in one vertex and leave  $K_{3,3}$  plus an isolated vertex, which holds no third  $K_4$ .

The cases  $d = 0$ . Now every colour is a matching and  $q = 2r + 1$ , so (1.1) is an equality for every point: every vertex has degree three and any two points meet once. So every witness block is a triple, and the  $F_i$  partition the edges of  $K_u$ .

- $(5, 2)$ :  $qr = 55$  is not divisible by three.
- For  $u \leq 9$  a local graph has at most six edges: one colour is a matching of at most four edges, and with several colours each is a matching of at most two, since a third disjoint edge would meet no other-colour edge. Six edges occur only as the three perfect matchings of a  $K_4$ , and five only as  $K_4$  minus an edge.
- $(9, 6)$ : the 36 edges of  $K_9$  would split into six copies of  $K_4$ , which needs the degree 8 to be divisible by three.
- $(7, 4)$ : the 21 edges of  $K_7$  split into four parts of size at most six only as  $6 + 6 + 6 + 3$ ,  $6 + 6 + 5 + 4$  or  $6 + 5 + 5 + 5$ . Two edge-disjoint  $K_4$  leave  $K_{3,3}$  plus a vertex, which excludes the first two. In the third, fix the  $K_4$  on four vertices. Each copy of  $K_4$  minus an edge contains a triangle, so it takes one of the three edges among the other three vertices; it is then two of the four joined to two of the three, plus that edge. The three pairs taken among the three are distinct and meet, so edge-disjointness needs three disjoint pairs among four vertices, which is impossible.

This proves Theorem 1.  $\square$

## 2.4 Sharpness, and degree four

For odd  $n$  take points  $1, \dots, n$  and one vertex for each pair  $([2])$ . Then  $q = n$ ,  $r = n - 1$ , a cover is an edge cover of  $K_n$ , and  $\tau = (n + 1)/2$ , so  $4\tau = q + r + 3$ .

**Corollary 2.** *If  $J$  has maximum degree at most four, then  $4\tau(J) \leq q + r + 3$ .*

*Proof.* Take a maximal family of disjoint 4-point blocks,  $a$  of them. Their vertices cover  $4a$  points and the rest has maximum degree three, so  $\tau \leq a + (q - 4a + r + 3)/4$ .  $\square$

Peeling at degree five then gives  $|E(H)| \geq (15r - q - 15)/4$  for a remainder of size  $q$ . This improves the additive term in the corresponding step of [2], not its coefficient.

## 3. The equality case

In Sections 3 and 4 suppose  $|E(H)| = 3r - 3$  and  $\tau(H) = r$  with  $r \geq 5$ . We write *deleted hyperedges* for the hyperedges removed in the reduction.

### 3.1 The parameters

The maximum degree of  $H$  is four:

- a vertex of degree at least five, peeled first, would give  $|E(H)| \geq |E(J)| + 4(r - \tau(J)) + 1 \geq 3r - 2$  by Theorem 1;
- if every degree were at most three,  $3r - 3 \leq 2r + 1$  would force  $r \leq 4$ . (At  $r = 3$  and  $r = 4$  the bound  $3r - 3$  is attained.)

Take a maximum family of disjoint 4-point blocks,  $a$  of them. Their  $4a$  hyperedges form *groups* of four, each with its *group vertex*. Delete them and let  $J$  be the rest: maximum degree three and  $q = 3r - 3 - 4a$ . The  $a$  group vertices and a cover of  $J$  cover  $H$ , so  $\tau(J) \geq r - a$ , while Theorem 1 gives  $4\tau(J) \leq q + r + 3 = 4(r - a)$ . So Theorem 1 is an equality for  $J$ , and §2.3 forces  $u$  odd,  $\tau(J) = r - a$ ,  $r = u + t - 1$  and  $q = 3t + u$ . Consequently

$$u = 2a + 3, \quad t = r - 2a - 2, \quad d = u - t - 1 \geq 0, \quad r = 2u - 2 - d, \quad (3.1)$$

and (2.1) becomes  $S \geq ut$ . Write  $H_0 = (u - 1)/2$ .

When  $d = 0$  every vertex of a point of  $J$  has degree three in  $J$ ; as  $d$  grows, vertices of degree two and one are allowed.

### 3.2 Local bounds

**Lemma 4 (pointwise capacity).** *For every  $x \in U$ ,  $S_x \geq t$ . In one  $F_i$ , if  $x$  carries  $E$  edges in  $k$  colours, then  $E - k \leq d$ .*

*Proof.* The point  $x$  has exactly  $u - 1$  witnesses, so exactly  $r - (u - 1) = t$  other vertices. They meet at most  $2t$  points of  $W$ , and all  $3t$  must be met, so  $3t - S_x \leq 2t$ . The  $u - 1$  witnesses at  $x$  reach  $S_x$  distinct points of  $W$ , so the number of witnesses *wasted* at  $x$  is  $u - 1 - S_x \leq u - 1 - t = d$ . Edges of one colour at  $x$  reach one point, so  $E$  edges in  $k$  colours waste  $E - k$ .  $\square$

**Lemma 5.** *Every  $F_i$  with more than one colour has  $S_i \leq \max(12, d + 8)$  and  $S_i - e(F_i) \leq 6$ . A one-coloured  $F_i$  has  $S_i - e(F_i) \leq H_0$ .*

*Proof.* The first case of Lemma 1 gives  $\max(12, d + 8)$ . In the star case,  $F_i$  is a star at one vertex with  $E$  edges in  $k \leq 3$  colours, and Lemma 4 gives  $S_i = E + k \leq d + 2k \leq d + 6$ .

For  $S_i - e(F_i)$ : if a colour has two disjoint edges on  $Y$ , either every support lies in  $Y$  and each colour has  $s - e \leq 2$ , or there are at most two other-colour edges, contributing at most two, while the first colour has  $e \geq s - 2$ . Otherwise every colour is a star or a triangle, with  $s - e \leq 1$ . A one-coloured graph has  $s - e$  at most its number of components, at most  $H_0$ .  $\square$

### 3.3 Two augmentation bounds

Call a one-coloured  $F_i$  *large* if its matching number is at least four, and let  $m$  be the number of large ones.

**Lemma 6.**  $2m \leq t$ .

*Proof.* A large  $F_i$  uses one colour  $z_i$ , so the other two points of  $M_i$  are met by no witness. Every  $x \in U$  must therefore meet these  $2m$  points through its  $t$  non-witness vertices, and no such vertex meets two of them:

- within one  $M_i$ , its block and an edge of  $F_i$  avoiding  $x$  would replace  $M_i$  by two triples;
- across  $M_i$  and  $M_j$ , its block, an edge of  $F_i$  avoiding  $x$ , and an edge of  $F_j$  avoiding  $x$  and that edge (matching number four provides both) would give three triples in place of two.

Either way the family of triples was not maximum.  $\square$

**Lemma 7.**  $u \leq \max(12, d + 8)$ .

*Proof.* Suppose not. By Lemma 5 every multicoloured  $F_i$  has  $S_i < u$ , and a one-coloured one has  $S_i \leq u$ . Since  $S \geq ut$ , every  $F_i$  is one-coloured and spans  $U$ . Then  $2e - u \leq d$  gives  $e \leq (u + d)/2$ , so its matching number is at least  $u - e \geq (u - d)/2 > 4$ . So  $m = t$ , and Lemma 6 forces  $t = 0$ , whereas  $t = u - d - 1 \geq 8$ .  $\square$

### 3.4 The swap lemma

The groups come from a *maximum* family of 4-point blocks. That alone constrains every deleted hyperedge.

**Lemma 8 (swap).** *Let  $A \neq A'$  be deleted hyperedges of the same group. If  $w \in A$  and  $w' \in A'$  are vertices of degree three in  $J$ , their sets of points of  $J$  meet.*

*Proof.* The vertex  $w$  lies in  $A$  and in three points of  $J$ , so its block is  $\{A\} \cup T$  with  $T$  three points of  $J$ ; likewise  $\{A'\} \cup T'$ . Neither block meets another group. If  $T \cap T' = \emptyset$ , the two blocks are disjoint, and putting them in place of the group's own block gives  $a + 1$  disjoint 4-point blocks, against maximality.  $\square$

Call such a  $T$  a *triple* of  $A$ . A deleted hyperedge meets  $J$  through its triples and through vertices of degree two or one in  $J$ .

**Proposition 1.** *In the equality setting  $d \neq 0$ , and  $d \neq 1$  when  $r \geq 11$ .*

*Proof.* Fix a group of four deleted hyperedges.

$d = 0$ . By §2.3,  $q = 2r + 1$  and every vertex of a point of  $J$  has degree three in  $J$ . So a deleted hyperedge  $A$  meets the  $2r + 1$  points of  $J$  through at most  $r - 1$  triples. Fix one,  $T$ . If the other at most  $r - 2$  triples all met  $T$ , they would reach at most  $3 + 2(r - 2) = 2r - 1$  points; so  $A$  has a triple  $T'$  disjoint from  $T$ . By Lemma 8 every triple of a group-mate  $A'$  meets both  $T$  and  $T'$ , so it reaches at most one of the  $2r - 5$  points outside them. With at most  $r - 1$  triples,  $A'$  misses one, since  $r - 1 < 2r - 5$  for  $r \geq 5$ .

$d = 1$ . Now  $q = 2r$ , and (1.1) allows each point of  $J$  at most one vertex of degree two and none of degree one. So  $J$  has at most  $q/2 = r$  vertices of degree two, each in at most two deleted hyperedges: at most  $2r$  incidences with deleted hyperedges.

- A deleted hyperedge whose triples pairwise meet reaches at most  $1 + 2n_3$  points through its  $n_3$  triples and two through each other vertex, at most  $1 + 2(r - 1) < 2r$  in all. So every deleted hyperedge has two disjoint triples.
- For a group-mate  $A'$  with  $n_2$  vertices of degree two, the  $2r - 6$  points outside those two triples are reached at most once per triple and twice per vertex of degree two, so  $2r - 6 \leq (r - 1) + n_2$  and  $n_2 \geq r - 5$ .

Every deleted hyperedge is such a group-mate, so the  $4a \geq 4$  deleted hyperedges need at least  $4(r - 5)$  incidences, more than  $2r$  when  $r \geq 11$ .  $\square$

## 4. Theorem 2: excluding equality from $r = 16$

**Theorem 2 (proposed).**  $g(r) \geq 3r - 2$  for every  $r \geq 16$ .

#### 4.1 Traces of the deleted hyperedges

For a deleted hyperedge  $A$ , its *trace*  $L_A$  is the graph on  $U$  with an edge  $xy$  whenever  $w_{xy} \in A$ . Write  $e_A$ ,  $s_A$ ,  $\nu_A$  for its number of edges, its support and its matching number. Put

$$T = \sum_i e(F_i), \quad D = S - T, \quad M = \sum_A (u - s_A).$$

- A witness of degree two in  $J$  lies in at most two deleted hyperedges, and one of degree three in at most one, since degrees in  $H$  are at most four. So  $\sum_A e_A \leq u(u-1) - T$ .
- For  $x \in U$ , its  $t$  non-witness vertices have room for at most  $3t$  further hyperedges. They must meet the  $3t - S_x$  points of  $W$  missed by the witnesses, and every deleted  $A$  whose trace misses  $x$ . So at most  $S_x$  deleted hyperedges miss  $x$ , and  $M \leq S$ .

With  $\nu_A \geq s_A - e_A$  and  $4a = 2u - 6$  this gives

$$\sum_A \nu_A \geq u^2 - 5u - D. \quad (4.1)$$

**Lemma 9.** *Within one group there are no three disjoint edges from three different traces.*

*Proof.* Extend them to a matching of  $H_0$  edges of the complete graph on  $U$ , leaving one point  $x$ . The witnesses of these edges cover  $U$  except  $x$  and meet three hyperedges of the group; one vertex of  $x$  that lies in the fourth covers both. With the  $t$  vertices of the triples and the  $a - 1$  other group vertices this is a cover of  $H$  of size  $H_0 + 1 + t + a - 1 = r - 1$ .  $\square$

**Lemma 10.** *Three traces each with a matching of size four, and a fourth nonempty trace, contain three disjoint edges from three different traces.*

**Lemma 11.** *The same holds for four traces with matching numbers at least 3, 3, 4, 4.*

*Proof of both.* Fix size-four matchings  $P, Q$  in two of the traces and suppose no such triple exists. Call an edge  $e$  of another trace *blocked* if no two disjoint edges, one from  $P$  and one from  $Q$ , avoid it; every edge of the other traces is blocked.

*Step 1: each blocked edge sits on a four-cycle.* The edges of  $P$  and of  $Q$  avoiding a blocked edge must cross-intersect. Each side keeps at least two, and one edge cannot meet three disjoint edges, so exactly two survive on each side. They form a four-cycle, which is a component of  $P \cup Q$  (counting an edge twice if it is in both). Both endpoints of  $e$  lie in both supports, since  $e$  removes exactly two edges from each matching.

*Step 2: two disjoint four-cycles.* The other traces supply three disjoint blocked edges (a size-three matching for Lemma 11, size four for Lemma 10). Their six endpoints lie in both supports, and outside one such four-cycle there are at most four such vertices, so another blocked edge touches it. The four-cycle of that edge avoids it, so it is a second, disjoint four-cycle. Hence  $P \cup Q$  is two disjoint four-cycles on the same eight vertices, and every blocked edge is a diagonal of one of them.

*Step 3: a rainbow triple.* For Lemma 11, the two size-three matchings each consist of three of the four diagonals. The first has both diagonals of one cycle and the second at least one of them; take that one for the second, the other for the first, and an edge of  $P$  in the other cycle. For Lemma 10, the three size-four matchings are the three perfect matchings of each of the two copies of  $K_4$ . Take an edge  $f$  of the fourth trace. If  $f$  removes two vertices of one copy, the edge on its other two vertices and an edge of a different matching in the other copy avoid  $f$ . If  $f$  removes at most one

vertex of each copy, each copy keeps a triangle or all of  $K_4$ , so two edges of different matchings avoid  $f$ .  $\square$

Order the matching numbers of one group as  $\nu_1 \leq \nu_2 \leq \nu_3 \leq \nu_4$ . Greedy choice gives a rainbow triple when  $\nu_2 \geq 1$ ,  $\nu_3 \geq 3$  and  $\nu_4 \geq 5$ . With Lemmas 9–11, every group has  $\nu_2 = 0$ , or  $\nu_3 \leq 2$ , or all  $\nu_i \leq 4$  with neither  $\nu_1 \geq 1$ ,  $\nu_2 = \nu_3 = \nu_4 = 4$  nor  $\nu_1, \nu_2 \geq 3$ ,  $\nu_3 = \nu_4 = 4$ . So the sum over one group is at most  $\max(2H_0, H_0 + 6, 13)$ , and with (4.1)

$$D \geq \frac{u^2 - 6u - 3}{2} \quad (u \geq 15), \quad D \geq 39 \quad (u = 13). \quad (4.2)$$

#### 4.2 Large remainders: $u$ is at most 11

$u \geq 15$ . By Lemma 7,  $d \geq u - 8$  and  $t = u - d - 1 \leq 7$ . By Lemma 5,  $D \leq 7H_0 = 7(u - 1)/2$ . With (4.2) this gives  $u^2 - 13u + 4 \leq 0$ , false for  $u \geq 13$ .

$u = 13$ . Lemma 7 gives  $d \geq 5$ .

- If  $d \geq 6$  then  $t \leq 6$  and  $D \leq 6t \leq 36$ , against (4.2).
- If  $d = 5$  then  $t = 7$  and every  $S_i \leq 13$ , so  $S \geq 91$  forces every  $S_i = 13$ . A one-coloured  $F_i$  then has  $e \leq 9$  and matching number at least four, so it is large and  $S_i - e \leq 6$ . A multicoloured  $F_i$  cannot be the star case of Lemma 5 (which gives at most  $d + 6 = 11$ ), so  $S_i = 13 = d + 8$  is the extremal case of Lemma 1: two other-colour edges  $pa$ ,  $pb$ , and a first colour made of  $ab$  and six edges at  $p$ , so  $S_i - e = 13 - 9 = 4$ . Lemma 6 gives  $m \leq 3$ , so  $D \leq 6 \cdot 3 + 4 \cdot 4 = 34$ , against (4.2).

So  $u \leq 11$ .

#### 4.3 $u = 11$ : the anchor count

**Open pairs.** A pair  $(x, c)$  is *open* in a local graph  $F_j$  if no edge of colour  $c$  meets  $x$  and every edge of the other two colours does.

**Anchor value.** For a local graph  $F_i$  and  $x \in U$ , let  $P_x$  be the set of colours of  $F_i$  with no edge at  $x$ ; their points of  $M_i$  are not reached by the witnesses at  $x$ .

- A pair of colours in  $P_x$  is *forbidden* if  $F_i$  has an edge of the third colour avoiding  $x$ .
- A colour  $c \in P_x$  is *good* if, for every pair  $f$  of  $U$  that avoids  $x$  and is not an edge of  $F_i$ , some edge of  $F_i$  of a colour other than  $c$  avoids both  $x$  and  $f$ .

For a set  $R \subseteq P_x$  whose pairs are all forbidden, let  $g_x$  be  $|R|$ , minus the number of colours of  $P_x$  outside  $R$ , minus the number of colours of  $R$  that are not good; take the best  $R$ . The *anchor value* of  $F_i$  is  $G = \sum_x \max(g_x, 0)$ .

**Lemma 12 (demand).** *If  $F_i$  has anchor value  $G$ , the other local graphs have at least  $tu + G - S$  open pairs in all.*

*Proof.* Fix  $x$  and  $R$ . The  $t$  non-witness vertices of  $x$  must reach the  $3t - S_x$  points of  $W$  that its witnesses miss, among them the points of  $R$ . No vertex reaches two points of  $R$ : its block and an edge of the third colour avoiding  $x$  would replace  $M_i$  by two triples. So  $|R|$  distinct vertices reach the points of  $R$ , each reaching at most one further point, and the other  $t - |R|$  reach at most two each. The  $|R|$  vertices therefore reach at least  $t + |R| - S_x$  further points missed by the witnesses, of which at most one per colour of  $P_x$  outside  $R$  lies in  $M_i$ .

Let  $\{x, c, q'\}$  be the block of such a vertex, with  $c \in R$  good and  $q' \in M_j$ ,  $j \neq i$ . No edge of colour  $q'$  in  $F_j$  meets  $x$ , since the witnesses at  $x$  miss  $q'$ . If an edge  $f$  of  $F_j$  of another colour avoided  $x$ , goodness would give an edge  $e$  of  $F_i$  of a colour other than  $c$  avoiding  $x$  and  $f$ , and the blocks of  $\{x, c, q'\}$ ,  $e$  and  $f$  would be three disjoint triples in place of  $M_i$  and  $M_j$ . So  $(x, q')$  is open in  $F_j$ . Distinct  $q'$  give distinct pairs, and each colour of  $R$  that is not good spoils at most one. So at least  $t + g_x - S_x$  open pairs come from  $x$ . Summing over the  $x$  with  $g_x > 0$ , and using  $S_x \geq t$  for the rest, gives  $tu + G - S$ .  $\square$

### Three anchors.

- A properly three-coloured  $K_4$  on a set  $Y$  has  $G = 3(u - 4)$ : for  $x \notin Y$  all three colours are missing, every pair is forbidden, and every colour is good, since the other two colours form a four-cycle on  $Y$  and a pair  $f$  outside  $F_i$  meets  $Y$  in at most one point.
- A spanning one-coloured large  $F_i$  has  $G = 2u$ : for every  $x$  its two unused colours are missing, forbidden together, and good, since a matching of size four leaves an edge avoiding  $x$  and  $f$ .
- The graph  $Y_k$ , with edges  $pa$  and  $pb$  of two colours and a third colour made of  $ab$  and  $k$  edges at  $p$ , has  $G = 2(u - 3)$ .

**Lemma 13 (finite classification; code/verify\_anchor\_count.py).** *Let  $u = 11$  and  $1 \leq d \leq 4$ . Every local graph with more than one colour has at most  $12 - S_j$  open pairs, and every one with  $S_j \geq 11$  has anchor value at least 16. A one-coloured graph of support  $s$  has  $11 - s$  open pairs, plus two for each vertex common to all its edges.*

The one-coloured count is direct. The script checks every multicoloured local graph on eleven vertices that satisfies (2.2) and Lemma 4: those with  $S_j \geq 11$  are  $K_4$  and  $Y_4$ , and at  $d = 4$  also  $Y_5$ , with anchor values 21, 16 and 16.

**Proposition 2.** *In the equality setting,  $u = 11$  with  $1 \leq d \leq 4$  is impossible.*

*Proof.* Here  $t = 10 - d$ , every  $S_j \leq 12$  by Lemma 5, and  $S \geq 11t$ . So  $\sum_j (12 - S_j) \leq t \leq 9$ , and some  $S_j \geq 11$ . That  $F_j$  is an anchor with  $G \geq 16$ : by Lemma 13 if it has several colours; otherwise it spans  $U$  with at most  $(11 + d)/2 \leq 7$  edges, so its matching number is at least four and  $G = 22$ . By Lemma 12 the other local graphs carry at least  $11t + 16 - S$  open pairs.

By Lemma 13 a local graph carries at most  $12 - S_j$  open pairs, except a one-coloured one whose edges share a vertex. A star with  $k$  leaves has  $12 - S_j = 11 - k$  and carries one more; a single edge or an empty graph would need  $12 - S_j \geq 10 > t$ . A star needs  $11 - k \leq t$  and, by (2.2),  $k \leq d + 1$ , so  $k = d + 1$  and it uses the whole budget; at most one occurs. So the local graphs carry at most  $12t - S + 1$  open pairs, and  $11t + 16 \leq 12t + 1$  would need  $t \geq 15$ .  $\square$

## 4.4 Proof of Theorem 2

Suppose  $r \geq 16$ ,  $|E(H)| = 3r - 3$  and  $\tau(H) = r$ . By §4.2,  $u \leq 11$ , and  $r = 2u - 2 - d$  by (3.1).

- If  $u \leq 9$ , then  $r \leq 16 - d$ , so  $r = 16$ ,  $u = 9$ ,  $d = 0$ , excluded by Proposition 1.
- If  $u = 11$ , then  $d = 20 - r \leq 4$ :  $d = 0$  is excluded by Proposition 1, and  $1 \leq d \leq 4$  by Proposition 2.

So  $|E(H)| \geq 3r - 2$  by Corollary 1.  $\square$

## 4.5 The remaining finite case

At  $r = 15$  the parameter choices are  $(u, d, t) = (9, 1, 7)$  and  $(11, 5, 5)$ . Proposition 1 excludes the first. The second remains open: at  $d = 5$  multicoloured stars can have support  $u = 11$  with zero anchor value, and the count of Lemma 13 no longer applies. So the argument stops at  $r \geq 16$  and does not settle  $7 \leq r \leq 15$ .

## 5. Verification

- Theorem 1 on every configuration with at most six hyperedges, and at low degree up to nine, and the sharpness family: `code/verify_residual_bound.py`.
- Lemmas 1, 3 and 5, the  $d = 0$  structure of §2.3, and the  $u = 13, d = 5$  step: `code/verify_local_graphs.py`.
- Lemmas 10 and 11 on 8, 9 and 10 vertices: `code/verify_rainbow_lemmas.py` and `code/verify_rainbow.c`.
- Lemma 13, the anchor values, and the failure at  $d = 5$  in §4.5: `code/verify_anchor_count.py`.
- The arithmetic of §2.3, §3.4 and §4: `code/verify_p5_arithmetic.py`.

The runs are recorded in `results/p5_checks.txt`. The steps that are not finite have been read but not independently reviewed.

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